NYC Airbnb Price Prediction (Linear Regression)

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Model Evaluation Metrics	
R-Squared (R2) Score:	**0.4868 (48.7%)**
Mean Absolute Error (MAE):	**$38.32**
Root Mean Squared Error (RMSE):	**$52.54**

Dataset Summary	
Source:	Kaggle NYC Airbnb 2019
Raw Listings:	48,895 rows
Consumer Listings (95th %):	45,748 rows
Train / Test Split:	36,598 / 9,150 (80/20)

EXECUTIVE SUMMARY FOR BUSINESS STAKEHOLDERS

> **What these numbers mean in plain English:**
> An R2 score of 0.487 means that our linear regression model explains approximately 48.7% of the total variation in nightly NYC Airbnb rates using only basic listing features like borough, room type, coordinates, and review counts. The Mean Absolute Error (MAE) of $38.32 means that on average, our predicted price is within $38 of the host's actual listing price.

KEY PIPELINE ENGINEERING DECISIONS

- **Data Cleaning:** I dropped the last_review date column because it had over 10,000 missing values. I noticed that reviews_per_month had missing values only when the listing had zero total reviews, so I filled those missing values with 0.0 rather than dropping listings. Identifier columns like id and host name were removed.
- **Outlier Treatment:** Luxury penthouses priced up to $10,000/night and minimum stays of 365+ nights heavily distort linear regression. Using the 95th percentile rule, I capped prices at $355.00 and minimum nights at 30, retaining 93.56% of typical consumer listings.
- **Avoiding the Curse of Dimensionality:** The dataset has 221 specific neighborhoods and 5 broad borough groups. One-hot encoding 221 neighborhoods would add over 220 sparse columns, causing the curse of dimensionality and overfitting. I dropped neighbourhood and kept neighbourhood_group with room_type, using drop_first=True to avoid the dummy variable trap.

MODEL PERFORMANCE: CHEAP VS EXPENSIVE LISTINGS

The model performs significantly better on budget listings ($40 to $150) than on premium listings ($200 to $355). Linear regression minimizes squared errors by finding the average trendline. Because over 75% of New York Airbnb listings are budget and mid-tier apartments in Brooklyn, Queens, and Upper Manhattan, the model weights prioritize this central density.

For luxury listings, price variation depends heavily on non-linear factors like skyline views, interior design, high end amenities, and seasonal demand. Since our dataset does not contain these details, our linear model predicts closer to the borough average, leading to noticeable underprediction on luxury properties.